



main.c...

Output



```
1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4  int main() {
5      int N = 4;
6      cout << "N = " << N << ":" << endl
           << endl;
7
8      // Nested loops to print the
           matrix
9      for (int i = 0; i < N; i++) {
           // Outer loop for rows
10     for (int j = 0; j < N; j++) { //
           Inner loop for columns
11     if (i == j || (i + j) % 2 == 0)
           {
12         cout << setw(3) << 1;
13     } else {
14         cout << setw(3) << 0;
15     }
16     }
17     cout << endl; // Moves to the
           next line after each row
18 }
19
20 return 0;
21 }
22 |
```

Run



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N = 4:

```
1 0 1 0
0 1 0 1
1 0 1 0
0 1 0 1
```

=== Code Execution Successful ===



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Output



```
1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4  int main() {
5      int totalPatients = 4;
6      int totalChecks = 12; // vitals
        checked every 2 hours across a
        24-hour shift
7      int p, c;
8      int hr;           // simulated
        heart rate for this patient
        /check
9      int alarmCount;   // counts
        alarms for the current patient
10
11     // ----- TODO 1: Print a top
        border -----
12     int borderLength = (totalChecks *
        4) + 12;
13     for (int i = 0; i < borderLength;
        i++) {
14         cout << "-";
15     }
16     cout << endl;
17
18     // ----- TODO 2: Print a
        header row of check numbers
        -----
19     for (int i = 0; i < borderLength; i++) {
```

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```
-----  
19     cout << setw(11) << left << " ";  
20     for (int i = 1; i <= totalChecks;  
21         i++) {  
22         cout << "C" << setw(3) << left  
23             << i;  
24     }  
25     cout << endl;  
26  
27     // Main nested loop structure  
28     for (p = 0; p < totalPatients; p  
29         ++){  
30  
31         alarmCount = 0;  
32         cout << "Patient " << p + 1 <<  
33             "<: "< ;  
34  
35         // ----- TODO 3: Inner  
36             loop over checks  
37             -----  
38         for (c = 0; c < totalChecks; c  
39             ++){  
40  
41             // Formula provided in the manual  
42             hr = 65 + (p * 8) + (c % 5) * 4  
43                 - ((c == 0 && p == 0) ? 10 :  
44                 0);  
45  
46             // ----- TODO 4: Flag  
47                 abnormal readings -----
```

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```
0);  
35 // ----- TODO 4: Flag  
    abnormal readings -----  
36 if (hr < 60 || hr > 100) {  
37     cout << hr << "(A) ";  
38     alarmCount++;  
39 } else {  
40     cout << setw(4) << left << hr;  
41     }  
42 }  
43 // ----- TODO 5: Report  
    the alarm count -----  
44 cout << "-> Total Alarms: " <<  
    alarmCount;  
45 cout << endl;  
46 }  
47  
48 // ----- TODO 6: Print a  
    bottom border -----  
49 for (int i = 0; i < borderLength;  
    i++) {  
50     cout << "-";  
51 }  
52 cout << endl;  
53  
54 return 0;  
55 }  
56
```

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```

-----
          C1  C2  C3  C4  C5  C6  C7  C8
          C9  C10 C11 C12
Patient 1: 55(A) 69  73  77  81  65  69  73
          77  81  65  69  -> Total Alarms: 1
Patient 2: 73  77  81  85  89  73  77  81
          85  89  73  77  -> Total Alarms: 0
Patient 3: 81  85  89  93  97  81  85  89
          93  97  81  85  -> Total Alarms: 0
Patient 4: 89  93  97  101(A) 105(A) 89  93
          97  101(A) 105(A) 89  93  -> Total
          Alarms: 4
-----

```

```

=== Code Execution Successful ===

```